MILITARY SPECIFICATION SHEET

CHAFF PACKAGE (TYPE RR-153A/AL)

This specification is approved for use by the Naval Air Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the chaff package described herein shall consist of this document and the latest issue of Specification MIL-C-85002.

TABLE I. Chaff dipole lengths and counts 1/

Dipole Cut No.	Dipole Length (Inches)	No. of Dipoles Per Chaff Cut (±10%)	No. of Cuts Per Chaff Package	No. of Dipoles Per Chaff Package (±10%)
1	0.55 ± 0.04	750,000	3	2,250,000
2	0.63 ± 0.04	750,000	2	1,500,000
3	1.07 ± 5%	750,000	1	750,000
4	1.84 ± 5%	750,000	2	1,500,000
5	1.61 ± 5%	750,000	1	750,000

1/ The number of dipoles per chaff cut and package is reference information.

APPLICABLE DOCUMENTS:

Military:

MIL-R-60346 Roving, Glass, Fibrous

REQUIREMENTS:

Item description: Aluminum coated glass fiber dipoles.

Detail requirements:

Components: Dipoles and non-chaff in accordance with DL 1050AS137.

Chaff dipoles: See table I.

Dipole material: Glass monofilament coated continuously with aluminum.

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Glass monofilament: Type E glass marbles; made from a composition similar to that delineated in MIL-R-60346.

Glass monofilament diameter: .95 mil maximum.

Aluminum coating: 99 percent or purer aluminum shall cover length continuous and around circumference at no less than 0.06 mil thickness.

Slip coating: Applied to dipole material in accordance with MIL-C-85002.

Weight, chaff only: 264 grams gross minimum.

Assembly:

Chaff package: Nine chaff cuts and non-chaff material assembled in accordance with DL 1050AS137.

Wrapping of dipoles: In order to facilitate packaging, dipoles may be wrapped using paper as specified on drawing 1050AS137. If the wrap is used and is greater than 1 1/2 turns, slit paper longitudinally to facilitate deployment of the dipoles. Chaff or wrap appearance shall not be degraded by slitting process.

Weight, Chaff package:

Gross weight of individual chaff package: 295 grams minimum.

Quality assurance provisions:

Radar requirements: Use M-33 radar with 1.5-2 degree beamwidth vertically polarized antenna.

Frequency: 9.060 ± .453 GHz

PRF: 1000 ± 100 pps

Pulsewidth: .25 us

Radar cross-section test procedures:

Aircraft operation: In accordance with figure 3 of MIL-C-85002 and the flight profile data of table II.

TABLE II. Flight profile test data.

Flight requirement	Value
Aspect Angle	5° -20°
Aircraft	180 kts. TAS
Altitude	3,500 feet
Dispense Command	10,000 yards
Standby Command	11,000 yards

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Data gathering: For acceptable data in gates 12 through 15, range in yards divided by sine of aspect angle shall be equal to or greater than 30,000.

Standby command: See table II.

Dispense command: See table II.

Data from gates 1 through 15: Spaced 200 nanoseconds in time.

Full scale value: $30m^2$

Data processing:

Radar resolution cells: 200 to 1300 feet behind the aircraft.

Data selection:

Running average: 175 consecutive pulse-to-pulse RCS values.

Correction factors:

Antenna gain: Antenna pattern as specified in MIL-STD-2071 or fourth order polynominal.

Fourth order polynominal for M-33 radar: $14.8242X^4 - 18.7422X^3 + 8.86719X^2 - 1.36035X + 1.0220$ (where X = angle off boresite in degrees).

Radar cross-section (RCS): Measured 500 through 800 feet behind aircraft shall meet or exceed 15 square per meters per gate.

Average of 10 or more units: Must equal or exceed 19 square meters per gate.

Preparing Activity
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